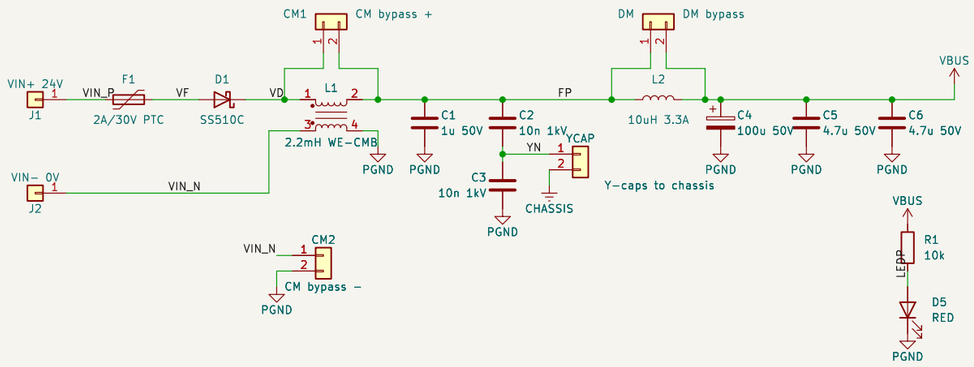
1. INPUT + SWITCHABLE EMI FILTER (J1/J2 to the two 5 uH LISNs)

CM1+CM2 fitted = CM choke bypassed YCAP = Y-caps to chassis DM = DM choke bypassed



6. TEST POINTS

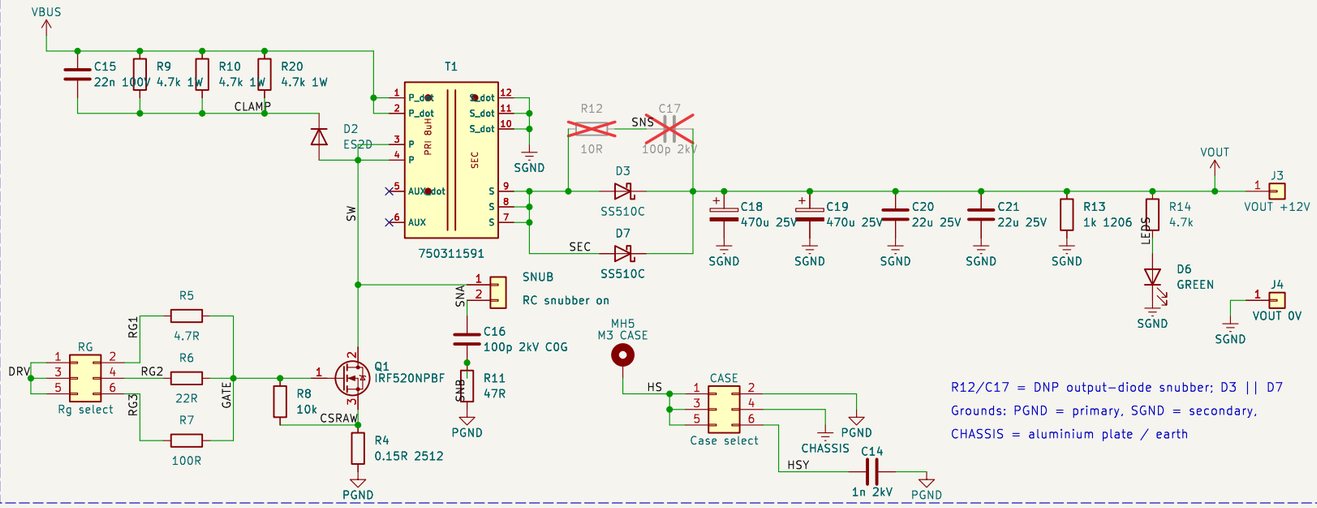
DRAIN SW · GATE GATE · CS CSRAW · HS HS · VBUS VBUS · VOUT VOUT · PGND1 PGND · PGND2 PGND · SGND SGND · CHAS CHASSIS

3. FLYBACK POWER STAGE + OUTPUT (T1 Wurth 750311591, fsw 154 kHz)

RG: 1-2 = 4.7R, 3-4 = 22R, 5-6 = 100R (fit one) SNUB = drain RC snubber

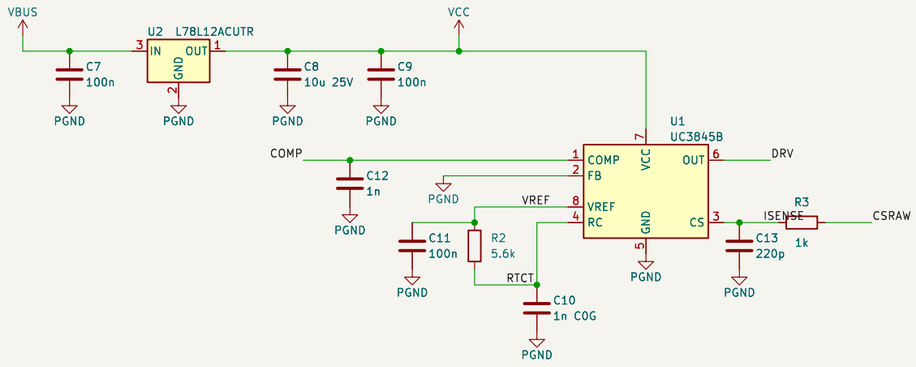
CASE (Q1 heatsink = Al case, via MH5): open = floating, 1-2 = PGND, 3-4 = CHASSIS, 5-6 = 1nF to PGND

Q1 on the PCB bottom, tab on the Al case floor (Sil-Pad 400, ~30 pF tab-to-case)



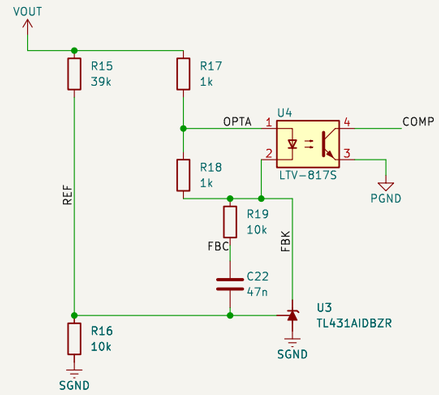
2. UC3845 CONTROLLER

fosc = 1.72/(R2*C10) = 307 kHz, fsw = 154 kHz; Ilim = 1V/R4 = 6.7 A



4. SECONDARY FEEDBACK (TL431 + opto)

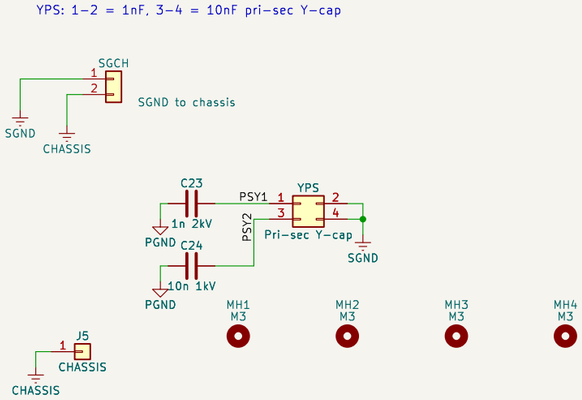
Vout = 2.495 V x (1 + R15/R16) = 12.2 V



5. ISOLATION / CHASSIS KNOBS

SGCH = SGND to chassis (earthed output)

YPS: 1-2 = 1nF, 3-4 = 10nF pri-sec Y-cap



7. DATASHEETS (click a line to open the datasheet; symbol properties also hold 'Datasheet' and 'Product page')

J1,J3 (VIN+ 24V): Cliff FCR7350R
J2,J4 (VIN- 0V): Cliff FCR7350B
F1 (2A/30V PTC): BSMD1812-200-30V
D1,D3,D7 (SS510C): SS510C
L1 (2.2mH WE-CMB): Wurth 744822222
CM1,CM2,+4 (CM bypass +): PZ254V-11-02P
C1 (1u 50V): CL31B105KBHNNNE
C2,C3,C24 (10n 1kV): CC1206KKX7RCBB103
L2 (10uH 3.3A): SWPA8040S100MT
C4 (100u 50V): MVK 50V100 M 10*10
C5,C6 (4.7u 50V): CL32B475KBUYNNE
R1,R8,+2 (10k): 0603WAF1002T5E
D5 (RED): NCD0805R1

U2 (L78L12ACUTR): L78L12ACUTR
C7,C9,C11 (100n): CC0603KRX7R9BB104
C8 (10u 25V): CL21A106KAYNNNE
U1 (UC3845B): UC3845BD1013TR
R2 (5.6k): 0603WAF5601T5E
C10 (1n C0G): CL10C102JB8NNNC
C12 (1n): CL10B102KB8NNNC
R3,R17,R18 (1k): 0603WAF1001T5E
C13 (220p): CL10B221KB8NNNC
R4 (0.15R 2512): HoLR2512-150mR-1%
RG,CASE (Rg select): PZ254V-12-6P
R5 (4.7R): 0805W8F470KT5E
R6 (22R): 0805W8F220JT5E

R7 (100R): 0805W8F1000T5E
Q1 (IRF520NPBF): IRF520NPBF
C14,C23 (1n 2kV): CC1206KKX7RDBB102
T1 (750311591): Wurth 750311591
D2 (ES2D): ES2D
C15 (22n 100V): CC0805KRX7R0BB223
R9,R10,R20 (4.7k 1W): 25121WJ0472T4E
C16,C17 (100p 2kV C0G): TCC1206C0G101J202DT
R11 (47R): 0805W8F470JT5E
R12 (10R): 1206W4F100JT5E
C18,C19 (470u 25V): RVT1E471M1010
C20,C21 (22u 25V): CL32A226KAJNNNE
R13 (1k 1206): 1206W4F1001T5E

R14 (4.7k): 0603WAF4701T5E
D6 (GREEN): KT-0805G
R15 (39k): 0603WAF3902T5E
U3 (TL431AIDBZR): TL431AIDBZR
C22 (47n): CL10B473KB8NNNC
U4 (LTV-817S): LTV-817S
YPS (Pri-sec Y-cap): PZ254V-12-4P
J5 (CHASSIS): Cliff FCR7350G
DRAIN,GATE,+4 (SW): Keystone 5002
PGND1,PGND2,+2 (PGND): Keystone 5001

Generated with Claude; verify before ordering

TU/e EPE – 5SWB0 Advanced Power Electronics

Sheet: /
File: emc_flyback.kicad_sch

Title: EMC teaching demo – 24V/12V flyback with EMC knobs

Size: A3	Date: 2026-10-03	**Rev: D**
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